

CORNELL TECH

**Cornell Tech Presents:
Builders and Disruptors in the AI Era**

Ann-Kareen Gedeus

**Ph.D. Student in Information Science
Advised by Assistant Professor Angelique Taylor**

Talk to Me Kindly: Designing AI Companions to Foster Positive Self-Talk in Young Adults

Ann-Kareen Gedeus | 2nd-Year Information Science PhD Student
Cornell University | Artificial Intelligence and Robotics Lab (AIRLab)

Advisor: Angelique Taylor

Talk: Builders and Disruptors in the AI Era

ChatGPT encouraged Adam Raine's suicidal thoughts. His family's lawyer says OpenAI is liable

More families sue Character.AI developer, alleging app played a role in teens' suicide and suicide attempt

SEP 16, 2025 ▾

By  Hadas Goldfarb

‘You’re not rushing. You’re just ready:’ Parents say ChatGPT encouraged son to kill himself

UPDATED NOV 20, 2025 ▾

By Rob Kuznia,  Allison Gordon,  Ed Lavandera

Research Problem

Young adults (age 18-25) face distinct stressors: lack of purpose, loneliness [6], uncertainty about the future [7], and socioeconomic/socio-political instability [8].

Their mental illness rates are more than 2x those of teenagers and are highest amongst all adult age groups [8,9].



Young adults often engage in negative self-talk (NST). NST occurs across age groups [1–3], yet young adults ruminate significantly more than older adults [4].

NST is a critical determinant of mental health in young adulthood [5].

Young adulthood is a formative period, and difficulties here can produce long-term effects into adulthood [3,10]

NST

*"I'm not confident. I
always mess up and
disappoint people."*

PST

*I've made mistakes
before, but they don't
define me. I can learn
and grow into who I
want to be"*

Prior Work

Many young adults avoid seeking help for PST due to stigma and preference for self-reliance [11].

This leads to increased use of AI tools (e.g., ChatGPT) as informal mental health supports [11-14].

- AI systems can deliver context-aware responses and imitate human-human conversation [15,16].

They have been used to support anxiety and depression [17], as mental health life coaches [18], and for journaling support [19].

Recent work calls for tailored, transparent, and context-aware support [20-22].

Human-computer Interaction (HCI) research has introduced VR avatars for self-talk practice [23], chatbots that teach PST strategies to tutors [24], and tools designed for autistic users [25].

GAP: However, the unique self-talk needs of young adults remain insufficiently addressed.

Starting with the AI's verbal communication:

How do we design an AI's verbal communication to deliver more personalized, safe, and context aware solutions to young adults?

Research Questions

RQ1: *How does the lived experience of therapy promoting positive self-talk strategies for young adults look like, and how can these real-world insights inform the design of AI companions?*

RQ2: *How can those design recommendations for fostering positive self-talk in young adults be translated into actionable practices for implementing such AI companions?*

Therapists: Semi-structured Interviews

- (1) How are young adults unique compared to other age groups?
- (1) What are the unique needs of young adults as it pertains to positive self-talk?
- (1) What positive self-talk exercises/strategies therapist use in therapy with young adults?
- (1) How do therapist envision non-clinical and supplementary AI tools for young adult patients?

Qualitative Results

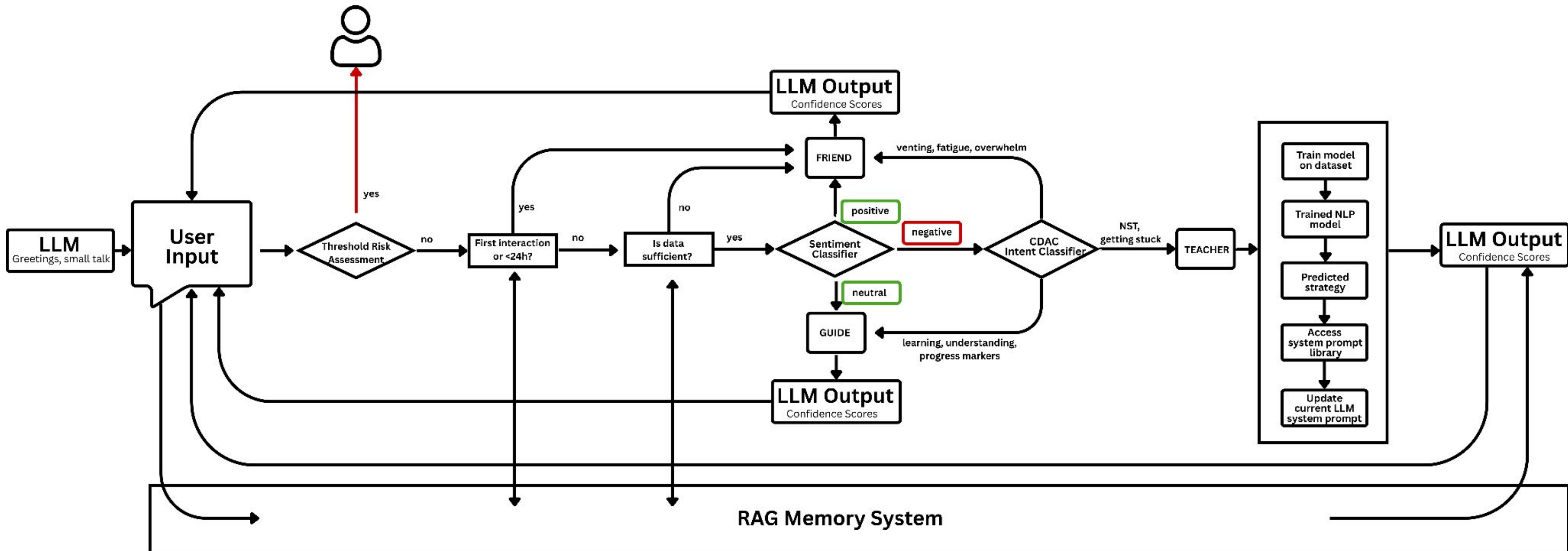
Key Finding 1

- Young adults' negative self-talk is identity-centered (self-image, peer comparison) — **Kim, Psychologist.**
- They hold fixed beliefs that resist change — **Logan, Mental Health Counselor.**
- They intellectualize emotions, separating knowing vs. feeling — **Kim, Psychologist.**

Key Finding 2

- Risk is never zero, it's always on a spectrum — **Kim, Psychologist.**
- AI must flag red-alert statements of self-harm or harm to others — **Micky, Psychologist.**
- Clinical tools like PHQ-9 are unsuitable in a non-clinical AI context — **Eva, Clinical Social Worker.**

Current: Human AI Self-Talk (HAIST) Framework



AI Experts: Semi-structured Interviews

- (1) Determine if PST's underlying assumptions are compatible with current AI capabilities.
- (1) Examine the viability of implementing PST strategies within existing LLM pipelines.
- (1) Validate whether therapist-derived PST actually resonates with young adults behaviors in digital contexts.

Contributions

- (1) A holistic understanding of behaviors and strategies therapists use to foster positive self-talk in young adults;
- (2) Design recommendations on how to design an AI companion promoting positive self-talk specifically to young adults; and
- (3) A conceptual framework called HAIST with actionable design guidelines for implementing PST support in non-clinical AI companions.

The MEER system for emotion regulation

Virtual Environment
Restorative design space:
initial openness, gamified
reinforcement over time



Metaverse

Ubiquitous access: across
devices and media



AI Companion

Supportive, pet-like avatar
fostering comfort and safe,
positive self-talk

Acknowledgements

Multidisciplinary team of students, faculty, and healthcare educators and professionals.

We thank NSF funding this exciting research.



Email: ag2637@cornell.edu

Website: <https://www.annkareengedeus.com/>

References

- [1] Calvete, E., & Cardenoso, O. (2002). Self-talk in adolescents: Dimensions, states of mind, and psychological maladjustment. *Cognitive Therapy and Research*, 26(4), 473-485
- [2] Ronan, K. R., & Kendall, P. C. (1997). Self-talk in distressed youth: States-of-mind and content specificity. *Journal of Clinical Child Psychology*, 26(4), 330-337.
- [3] Zimmermann, P., & Iwanski, A. (2014). Emotion regulation from early adolescence to emerging adulthood and middle adulthood: Age differences, gender differences, and emotion-specific developmental variations. *International journal of behavioral development*, 38(2), 182-194.
- [4] Emery, L., Sorrell, A., & Miles, C. (2020). Age differences in negative, but not positive, rumination. *The Journals of Gerontology: Series B*, 75(1), 80-84.
- [5] Grzybowski, J., & Brinthaup, T. M. (2022). Trait mindfulness, self-compassion, and self-talk: A correlational analysis of young adults. *Behavioral Sciences*, 12(9), 300.
- [6] Qirtas, M. M., Zafeiridi, E., White, E. B., & Pesch, D. (2023). The relationship between loneliness and depression among college students: Mining data derived from passive sensing. *Digital Health*, 9, 20552076231211104.
- [7] Schweizer, S., Lawson, R. P., & Blakemore, S. J. (2023). Uncertainty as a driver of the youth mental health crisis. *Current opinion in psychology*, 53, 101657.
- [8] Weissbourd, R., Batanova, M., McIntyre, J., Torres, E., Irving, S., Eskander, S., & Bhai, K. (2023). On Edge Understanding and Preventing Young Adults' Mental Health Challenges. https://static1.squarespace.com/static/5b7c56e255b02c683659fe43/t/6537db8894f0802b6480d38e/1698159503140/On+Edge_FINAL.pdf
- [9] National Institute of Mental Health. (2024a, September). Mental Illness. National Institute of Mental Health. <https://www.nimh.nih.gov/health/statistics/mental-illness>
- [10] Schulenberg, J. E., Sameroff, A. J., & Cicchetti, D. (2004). The transition to adulthood as a critical juncture in the course of psychopathology and mental health. *Development and psychopathology*, 16(4), 799-806.
- [11] Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): a randomized controlled trial. *JMIR mental health*, 4(2), e7785.
- [12] Ishikawa, A., Rickwood, D., Bariola, E., & Bhullar, N. (2023). Autonomy versus support: self-reliance and help-seeking for mental health problems in young people. *Social psychiatry and psychiatric epidemiology*, 58(3), 489-499.
- [13] Siddals, S., Torous, J., & Coxon, A. (2024). "It happened to be the perfect thing": experiences of generative AI chatbots for mental health. *npj Mental Health Research*, 3(1), 48
- [14] Alanezi, F. (2024). Assessing the effectiveness of ChatGPT in delivering mental health support: a qualitative study. *Journal of multidisciplinary healthcare*, 461-471.
- [15] Stade, E. C., Stirman, S. W., Ungar, L. H., Boland, C. L., Schwartz, H. A., Yaden, D. B., ... & Eichstaedt, J. C. (2024). Large language models could change the future of behavioral healthcare: a proposal for responsible development and evaluation. *NPJ Mental Health Research*, 3(1), 12.
- [16] Hua, Y., Na, H., Li, Z., Liu, F., Fang, X., Clifton, D., & Torous, J. (2025). A scoping review of large language models for generative tasks in mental health care. *npj Digital Medicine*, 8(1), 230.
- [17] Tao, Y., Yang, M., Shen, H., Yang, Z., Weng, Z., & Hu, B. (2023, December). Classifying anxiety and depression through LLMs virtual interactions: a case study with ChatGPT. In 2023 IEEE international conference on bioinformatics and biomedicine (BIBM) (pp. 2259-2264). IEEE.
- [18] Meywirth, S. (2024, May). Designing a large language model-based coaching intervention for lifestyle behavior change. In *International Conference on Design Science Research in Information Systems and Technology* (pp. 81-94). Cham: Springer Nature Switzerland.
- [19] Liu, Y., Chen, Z., Xie, X., Hu, Y., Zhang, L., Xu, D., & Wang, Y. (2024, September). MoodMonster: Integrating AI and Art for Emotional Journaling. In 2024 Artificial Intelligence x Humanities, Education, and Art (AIxHEART) (pp. 92-97). IEEE.
- [20] Moylan, K., & Doherty, K. (2025). Expert and Interdisciplinary Analysis of AI-Driven Chatbots for Mental Health Support: Mixed Methods Study. *Journal of Medical Internet Research*, 27, e67114.
- [21] Boucher, E. M., Harake, N. R., Ward, H. E., Stoeckl, S. E., Vargas, J., Minkel, J., ... & Zilca, R. (2021). Artificially intelligent chatbots in digital mental health interventions: a review. *Expert review of medical devices*, 18(sup1), 37-49.
- [22] Balcombe, L. (2023, October). AI chatbots in digital mental health. In *Informatics* (Vol. 10, No. 4, p. 82). MDPI.
- [23] Zisquit, M., Shoa, A., Oliva, R., Perry, S., Spanlang, B., Klomek, A. B., ... & Friedman, D. (2025). AI-Enhanced Virtual Reality Self-Talk for Psychological Counseling: Formative Qualitative Study. *JMIR Formative Research*, 9(1), e67782.
- [24] Thomas, D. R., Lin, J., Bhushan, S., Abboud, R., Gatz, E., Gupta, S., & Koedinger, K. R. (2024, July). Learning and ai evaluation of tutors responding to students engaging in negative self-talk. In *Proceedings of the Eleventh ACM Conference on Learning@ Scale* (pp. 481-485).
- [25] Carik, B., Izaac, V. V., Ding, X., Scarpa, A., & Rho, E. H. (2025, April). Reimagining Support: Exploring Autistic Individuals' Visions for AI in Coping with Negative Self-Talk. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (pp. 1-30).